

Expressing Discrete Geometry using the Conformal Model

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1. Introduction to Discrete Geometry

- **Discrete Geometry Focus:** Representation and transformation of geometric objects in discrete spaces (pixels, voxels).
- **Historical Foundation:** Analytical description of 2D discrete lines by J.-P. Réveillès (1988).
- **Core Challenge:** Describing classical geometric objects (lines, circles, planes, spheres) in a discrete context across arbitrary dimensions.
- **Proposed Solution:** Use the *conformal model* from geometric algebra to unify discrete primitive definitions.
- **Key Advantage:** Minimal expressions and algorithms valid across all dimensions.

Visual: Diagram illustrating transition from continuous to discrete geometry — pixels/voxels approximating smooth curves.

2. Conformal Model Basics

Extends Euclidean space $\mathbb{E}^n$ by adding two basis vectors e_+ and e_- .

[Aveneau2012-Equation0a] Origin point

$$n_o = \frac{1}{2}(e_- - e_+)$$

[Aveneau2012-Equation0b] Infinity point

$$n_\infty = e_- + e_+$$

[Aveneau2012-Equation0c] Conformal mapping of point p

$$F(p) = p + \frac{1}{2}p^2 n_\infty + n_o$$

3. Discrete Spheres — Definition

A discrete sphere is defined by center c , radius r , and width d .

[Aveneau2012-Equation1] General Discrete Sphere

$$S = \{p \in \mathbb{Z}^n \mid (r - d)^2 \leq |c - p|^2 < (r + d)^2\}$$

[Aveneau2012-Equation2] Conformal model — unit sphere at origin

$$S_1 \cdot F(p) = \frac{1}{2}(1 - p^2)$$

The sign of $\frac{1}{2}(1 - p^2)$ determines whether p lies inside or outside the sphere.

Visual: Graphical representation of a discrete sphere showing inner/outer boundaries defined by width d .

4. Discrete Spheres — Conformal Representation

[Aveneau2012-Equation3a] Outer boundary sphere

$$S_{c,r+d} = c - \frac{1}{2}(r+d)^2 n_\infty$$

[Aveneau2012-Equation3b] Inner boundary sphere

$$S_{c,r-d} = c - \frac{1}{2}(r-d)^2 n_\infty$$

[Aveneau2012-Equation4] Membership condition

$$S_{c,r+d} \cdot F(p) \leq 0 \quad \text{and} \quad S_{c,r-d} \cdot F(p) > 0$$

5. Discrete Hyperplanes in $\mathbb{R}^n$

A hyperplane is a sphere passing through n_∞ .

[Aveneau2012-Equation5] Discrete Hyperplane

$$H = \{p \in \mathbb{Z}^n \mid -d < \hat{n} \cdot p + \delta \leq d\}$$

[Aveneau2012-Equation6] Conformal representation

$$\pi = \hat{n} + \delta n_\infty$$

Visual: 2D discrete line with normal $\hat{n}$, offset δ , and discrete points within width d .

6. Discrete Rounds (k -Spheres)

A *round* is the intersection of $(n - k)$ hyperspheres in $\mathbb{R}^n$.

[Aveneau2012-Equation7] General Round

$$R = S_1 \wedge S_2 \wedge \cdots \wedge S_{n-k}$$

[Aveneau2012-Equation8] Discrete Round Membership

A point $p \in \mathbb{Z}^n$ belongs to the discrete round if it satisfies the membership conditions for *each* bounding sphere pair simultaneously.

7. Discrete Circles and Lines

[Aveneau2012-Equation9] Discrete Circle

$$C = S_1 \wedge S_2$$

Intersection of two conformal spheres.

[Aveneau2012-Equation10] Discrete Line

$$L = p_1 \wedge p_2 \wedge n_\infty$$

A line passes through two conformal points and the point at infinity.

Visual: Side-by-side discrete circle and line in 2D. Consider ganja.js or Manim for interactive/animated constructions.

8. Unified Framework — Summary

Object	Classical	Conformal	Ref.
Sphere	$(r-d)^2 \leq c-p ^2 < (r+d)^2$	$S = c - \frac{1}{2}r^2 n_\infty$	Eq. 1,3
Hyperplane	$-d < \hat{n} \cdot p + \delta \leq d$	$\pi = \hat{n} + \delta n_\infty$	Eq. 5,6
Round	$\bigcap$ of $(n-k)$ spheres	$R = S_1 \wedge \dots$	Eq. 7
Circle	$\bigcap$ of 2 spheres	$C = S_1 \wedge S_2$	Eq. 9
Line	Circle through ∞	$L = p_1 \wedge p_2 \wedge n_\infty$	Eq. 10

9. Conclusions and Future Work

- **Unification:** A single algebraic framework for all discrete primitives across arbitrary dimensions.
- **Minimality:** Compact, dimension-independent expressions.
- **Algorithmic Potential:** Generalized algorithms for discrete geometry operations.
- **Future Directions:**
 - Extension to conics and quadrics.
 - Efficient algorithm development from conformal representations.
 - Integration with computer graphics pipelines.
 - Discrete conformal transformations (rotations, translations, dilations).

Reference

[Aveneau2012] Aveneau, L., Andres, E., Mora, F. (2012). *Expressing Discrete Geometry using the Conformal Model*. AGACSE 2012, La Rochelle, France.